GM-2

Roshan Yadav
Roll No: 2311144[1]

[1] *Nuclear Physics and Instrumentation Laboratory, School of Physical Sciences, National Institute of Science Education and Research (NISER), Bhubaneswar, India*
Date of Experiment: 27/03/2026, 28/03/2026
Date of Submission: 02/04/2026

The interaction of beta radiation with matter was investigated using a Geiger–Müller counting system through measurements of absorption, backscattering, and bremsstrahlung production. The absorption of beta particles from Tl-204 and Sr-90 sources was studied using aluminium absorbers, and the corresponding ranges were determined as 0.0325 g/cm^2 and 0.0976 g/cm^2, yielding endpoint energies of 0.236 MeV and 0.361 MeV, respectively. Backscattering measurements showed saturation beyond a thickness of 0.15 mm, indicating a finite interaction depth for electron scattering. Bremsstrahlung radiation was examined for different material configurations, but the observed variations in intensity were comparable to statistical uncertainties, limiting definitive conclusions regarding material dependence. The results are broadly consistent with theoretical expectations of radiation-matter interaction, with deviations attributable to finite counting statistics and experimental constraints.

INTRODUCTION

The interaction of charged particles and electromagnetic radiation with matter forms a fundamental aspect of nuclear and radiation physics. Experimental investigation of these interactions provides insight into energy loss mechanisms, radiation transport, and detector response. The Geiger–Müller (G.M.) counter, owing to its high sensitivity and simplicity, serves as an effective tool for studying such phenomena.

In this work, three key processes are examined: (i) the absorption of beta particles in matter to determine their range and endpoint energy, (ii) the backscattering of beta particles from materials of varying thickness, and (iii) the production and attenuation of bremsstrahlung radiation. These experiments collectively illustrate the interaction of charged particles with matter, including ionization losses, scattering processes, and radiation emission due to deceleration.

The absorption of beta particles provides a method to estimate their maximum energy through range-energy relations. Backscattering studies reveal the dependence of scattering probability on material thickness and atomic properties. Bremsstrahlung, arising from the deceleration of charged particles in matter, highlights the conversion of kinetic energy into electromagnetic radiation. Together, these experiments provide a comprehensive understanding of radiation-matter interaction relevant to nuclear physics and detector applications.

THEORY

Absorption and Range of Beta Particles: Beta particles (electrons) lose energy primarily through ionization and excitation of atoms in the absorbing medium. As they traverse matter, their energy decreases continuously until they come to rest. The maximum distance traveled by beta particles in a material is defined as their *range*. Empirically, the range R (in g/cm^2) is related to the endpoint energy E_0 (in MeV) by

$$R = 0.52E_0 - 0.09,$$

as given in the manual (p. 36) :contentReference[oaicite:0]index=0.

Experimentally, the range is determined from the absorber thickness at which the count rate falls to approximately half its initial value or extrapolates to zero. By measuring the absorption curve (count rate versus absorber thickness), the endpoint energy of the beta spectrum can be estimated. This method relies on the continuous energy distribution characteristic of beta decay.

Backscattering of Beta Particles: When beta particles interact with matter, elastic and inelastic collisions with atomic electrons and nuclei can lead to scattering. A fraction of incident particles is scattered backward, a phenomenon known as backscattering. As described in the manual (p. 38), the probability of backscattering increases with atomic number and material thickness.

Initially, as the thickness of the scatterer increases, the number of backscattered particles rises due to increased interaction probability. However, beyond a certain thickness—known as the *saturation thickness*—the count rate becomes nearly constant, as most incident particles are either absorbed or scattered within the material. This thickness corresponds to a condition where additional material does not significantly increase the backscattering yield.

Bremsstrahlung Radiation: Bremsstrahlung ("braking radiation") is produced when a charged particle, such as an electron, is decelerated in the electric field of an atomic nucleus. This acceleration results in the emis-

sion of electromagnetic radiation with a continuous spectrum. As outlined in the manual (p. 41), the intensity of bremsstrahlung increases with the energy of the incident particle and the atomic number of the absorber material.

The production of bremsstrahlung becomes significant when high-energy beta particles interact with high-Z materials. Conversely, low-Z materials result in reduced bremsstrahlung yield. The emitted radiation can be subsequently absorbed depending on the material properties and thickness of the absorber. Thus, the observed count rate depends on both the generation and attenuation of bremsstrahlung within the experimental setup.

Radiation Attenuation: As radiation passes through matter, its intensity decreases due to absorption and scattering processes. For a homogeneous absorber, the attenuation can be described by an exponential law,

$$I = I_0 e^{-\mu x},$$

where μ is the linear attenuation coefficient and x is the thickness of the absorber. Although beta absorption deviates from a simple exponential due to energy loss and scattering, bremsstrahlung attenuation approximately follows this behavior for sufficiently thick absorbers.

Detector Response and Counting: The G.M. detector registers individual ionizing events as discrete pulses. The measured count rate is proportional to the incident radiation intensity, after correcting for background counts. Accurate measurements require subtraction of background radiation and consideration of detector limitations such as dead time.

These theoretical principles form the basis for analyzing absorption curves, estimating beta energies, understanding scattering mechanisms, and interpreting bremsstrahlung production in the present experiments.

EXPERIMENTAL SETUP AND PRINCIPLE

The experiments were performed using a Geiger–Müller (G.M.) counting system (GC601A/GC602A) coupled to an end-window G.M. detector mounted on a sliding optical bench. The detector was connected to the counting unit through coaxial cables and operated at a constant high voltage of approximately 500 V, corresponding to the plateau region of the detector characteristics. Radioactive sources and absorbers were positioned along the optical bench to ensure controlled geometry and reproducibility of measurements.

Beta Absorption and Range Measurement: To study the absorption of beta particles, radioactive sources (Tl-204 and Sr-90) were placed at a fixed distance (~ 3 cm) from the detector window. Aluminium absorber foils of known thickness were inserted between the source and detector. The absorber thickness was increased in small increments (typically 0.05 mm), and the corresponding count rates were recorded for a fixed counting time. Background radiation was measured separately and subtracted to obtain net count rates. The resulting absorption curve (count rate versus absorber thickness) was used to determine the effective range of beta particles and estimate their endpoint energy.

Backscattering of Beta Particles: The backscattering experiment was carried out by placing a beta source (Sr-90) and detector in a fixed geometry with a scatterer (aluminium foil) positioned adjacent to the source. A lead block was introduced between the source and detector to prevent direct radiation, ensuring that only backscattered particles contributed to the measured counts. The thickness of the scatterer was varied systematically, and the corresponding count rates were recorded. This configuration isolates the contribution of backscattered electrons and enables the study of the dependence of scattering yield on material thickness.

Bremsstrahlung Production and Attenuation: Bremsstrahlung radiation was investigated using combinations of absorber materials with different atomic numbers (Al, Cu, and Perspex). The beta source (Sr-90) was placed in proximity to the detector, and absorbers were arranged in different configurations (metal facing source, plastic facing source, etc.). Measurements were taken for each configuration to compare the intensity of bremsstrahlung radiation produced and transmitted through the absorber system. Since bremsstrahlung yield depends strongly on the atomic number of the material, these configurations allow qualitative and quantitative comparison of radiation generation and attenuation.

In all experiments, the counting system registers discrete pulses corresponding to ionizing events within the detector. The measured count rates were corrected for background radiation and used to infer physical quantities such as particle range, scattering probability, and radiation intensity. Care was taken to maintain fixed geometry, constant counting time, and stable operating voltage to ensure the reliability of the measurements.

RESULTS AND DISCUSSION

Beta Absorption and Range: The absorption curves for Tl-204 and Sr-90 are shown in Figs. 1 and 2. In both cases, the count rate decreases smoothly with increasing absorber thickness, consistent with continuous energy loss of beta particles through ionization and excitation processes.

Using the half-value method, the range for Tl-204 was determined to be $R_{\mathrm{Tl}} = 0.0325$ g/cm^2, corresponding to an endpoint energy of $E_{\mathrm{Tl}} \approx 0.236$ MeV. For Sr-90, a larger range of $R_{\mathrm{Sr}} = 0.0976$ g/cm^2 was

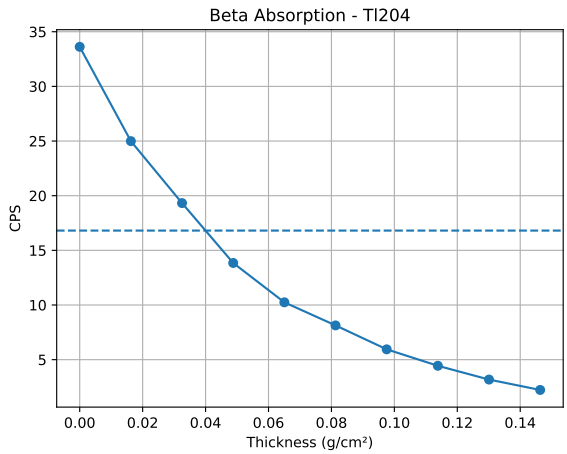

FIG. 1. Beta absorption curve for Tl-204 showing decrease of count rate with absorber thickness.

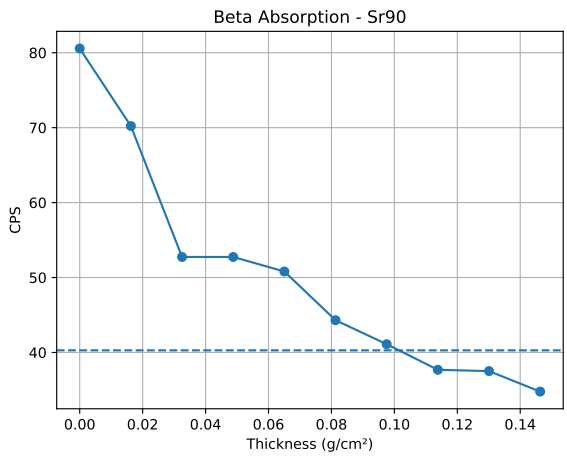

FIG. 2. Beta absorption curve for Sr-90 showing higher penetration compared to Tl-204.

obtained, yielding $E_{\mathrm{Sr}} \approx 0.361$ MeV :contentReference[oaicite:0]index=0.

The significantly larger range of Sr-90 confirms its higher beta energy. The absence of a sharp cutoff in both curves reflects the continuous beta spectrum. Minor deviations from smooth behavior may arise due to scattering and energy straggling within the absorber.

Backscattering of Beta Particles: The dependence of backscattered counts on absorber thickness is shown in Fig. 3.

The count rate initially increases with thickness, indicating an increased probability of scattering events. Beyond a thickness of approximately 0.15 mm, the count rate saturates :contentReference[oaicite:1]index=1, suggesting that additional material does not contribute significantly to further backscattering. This saturation arises because most incident beta particles are either absorbed or scattered within a finite interaction depth.

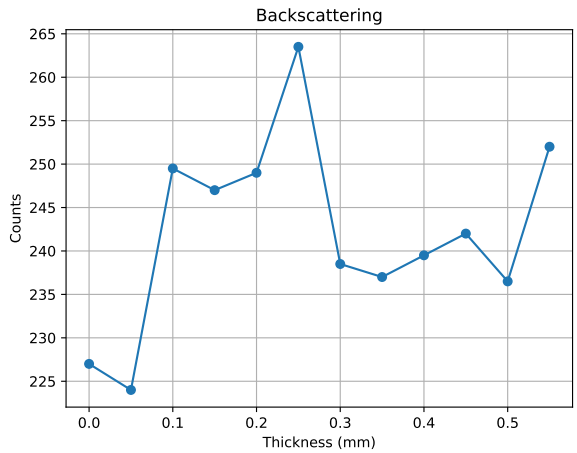

FIG. 3. Backscattering count rate as a function of aluminium thickness.

The relatively small variation in counts beyond saturation indicates that backscattering is predominantly a near-surface phenomenon, consistent with theoretical expectations.

Bremsstrahlung Radiation: The bremsstrahlung measurements were performed for three configurations: Al–Perspex, Cu–Perspex, and Cu–Al. The corresponding net count rates were found to be 1336.67, 1302.00, and 1294.33 counts respectively :contentReference[oaicite:2]index=2, with relative intensities close to unity.

In principle, bremsstrahlung production is expected to increase strongly with atomic number (Z), since the radiation intensity scales approximately as Z^2. Therefore, configurations involving copper (high-Z material) should produce higher radiation compared to aluminium or perspex (low-Z materials).

However, the observed differences between configurations are relatively small and lie within the experimental uncertainties. A key observation is the very large standard deviation in the measured counts (of order ~ 2000), which is comparable to or larger than the differences between configurations. This indicates that statistical fluctuations dominate the measurement, reducing the sensitivity to distinguish between different material effects.

Furthermore, the experiment inherently involves competing processes: while high-Z materials enhance bremsstrahlung production, they also increase absorption of the emitted photons. Thus, the detected intensity represents a balance between generation and attenuation. This interplay can partially obscure the expected dependence on atomic number.

Another contributing factor is the presence of background radiation and detector limitations, which further reduce the effective signal-to-noise ratio. As a result,

although the qualitative trend is consistent with theoretical expectations, the experimental data do not exhibit a strong or statistically significant dependence on material configuration.

Overall Discussion: The experiments collectively demonstrate key aspects of radiation-matter interaction. The beta absorption study confirms the continuous energy loss mechanism and enables estimation of particle energies. The backscattering experiment reveals saturation behavior associated with finite interaction depth. The bremsstrahlung experiment highlights the complexity arising from competing production and attenuation processes, as well as the limitations imposed by statistical uncertainties.

Code Availability: The complete data analysis pipeline used in this work is available at: `https://github.com/RoshanatNiser/Reports/blob/main/GM/GM_2.py`

Sources of Error

Several sources of uncertainty affect the accuracy and precision of the present measurements.

Statistical Fluctuations: Radioactive decay is inherently stochastic, and the number of counts follows Poisson statistics. The uncertainty in counting is therefore of order $\sqrt{N}$, which becomes significant for low count rates. In the bremsstrahlung experiment, the large standard deviation compared to the mean differences indicates that statistical fluctuations dominate the measurement.

Background Radiation: Background counts arising from cosmic rays and environmental radioactivity introduce systematic uncertainty. Any temporal variation in background radiation during measurements can lead to errors in net count estimation.

Detector Limitations: The finite dead time of the G.M. detector can lead to undercounting at higher count rates. Additionally, the detector efficiency is not uniform for all radiation energies, which may affect quantitative comparisons.

Thickness and Geometry Uncertainty: Errors in the measurement of absorber thickness and misalignment between the source, absorber, and detector can affect the observed count rates. Small deviations in geometry can lead to significant variations due to changes in effective solid angle.

Energy Loss and Scattering Effects: Beta particles undergo multiple scattering and energy straggling in the absorber, leading to deviations from ideal absorption behavior. This contributes to uncertainty in determining the range and endpoint energy.

Limited Measurement Range: In the absorption experiment, the full stopping region of beta particles was not completely reached, leading to an underestimation of the range and hence the endpoint energy, particularly for Sr-90.

CONCLUSION

The experiments successfully demonstrated key aspects of the interaction of beta radiation with matter. The absorption measurements confirmed the continuous energy loss of beta particles and enabled estimation of their effective range and endpoint energy. The higher range observed for Sr-90 relative to Tl-204 is consistent with its higher beta energy, although the estimated energy is lower than expected due to incomplete absorption measurements.

The backscattering study revealed a clear saturation behavior, indicating that electron scattering is predominantly governed by a finite interaction depth within the material. The observed saturation thickness is in agreement with theoretical expectations.

The bremsstrahlung experiment highlighted the interplay between radiation production and attenuation. Although high-Z materials are expected to enhance bremsstrahlung emission, the experimental results showed only marginal differences between configurations. This is attributed to large statistical fluctuations, as reflected in the high standard deviation of the measured counts, which limits the ability to resolve material-dependent effects.

Overall, the results are qualitatively consistent with theoretical predictions, while quantitative discrepancies arise primarily from statistical uncertainties, detector limitations, and the restricted range of measurements. The study underscores the importance of counting statistics and careful experimental design in nuclear radiation measurements.

[1] NISER Physics Laboratory Manual, *Nuclear Physics and Instrumentation Lab*, Experiments on G.M. Counter, Beta Absorption, Backscattering, and Bremsstrahlung.

[2] G. F. Knoll, *Radiation Detection and Measurement*, 4th ed. (Wiley, 2010).

[3] K. S. Krane, *Introductory Nuclear Physics* (Wiley, 1987).

[4] W. R. Leo, *Techniques for Nuclear and Particle Physics Experiments* (Springer, 1994).

[5] R. N. (Author), "Data Analysis Code for G.M. Experiments," GitHub repository: `https://github.com/RoshanatNiser/Reports/blob/main/GM/GM_2.py`